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Fire-rated modular duct assembly suitable for exhausting flammable or hazardous gases, vapours and other materials

Abstract

A fire-rated exhaust duct system comprising a modular configuration or structure. The fire-rated exhaust duct system comprises a plurality of exhaust duct sections. Each of the exhaust duct sections is configured to be joined or connected with other exhaust duct sections in the field or at an installation site, e.g. in building housing a kitchen facility or a laboratory facility to form longer sections or runs for exhaust duct system.

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Background/Summary

.Iadd.CROSS-REFERENCE TO RELATED APPLICATIONS .Iaddend. (1) .Iadd.Notice: More than one reissue application has been filed for the reissue of U.S. Pat. No. 9,557,071 ('071 patent), which applications include the present application, and Reissue application Ser. No. 16/261,258, filed on Jan. 29, 2019 ('258 application). The present application is a continuation reissue of the '258 application. The '258 application is a reissue of the '071 patent, and the '071 patent is a continuation of U.S. Pat. No. 9,074,788, filed on Jan. 6, 2012. The present application is also a reissue of the '071 patent..Iaddend.

.[.BACKGROUND OF THE INVENTION.].
.Iadd.FIELD OF THE INVENTION .Iaddend.
(1) The present application relates to duct assembly, and more particularly, to a fire-rated modular duct assembly suitable for exhausting flammable or hazardous gases, vapour and the like.
BACKGROUND OF THE INVENTION
(2) Many processes in commercial and industrial facilities generate flammable or hazardous gases, vapors or particles. The hazardous material must be captured at the source and transported or moved through the facility (e.g. building) to a location where the material can be discharged, e.g. directly into the atmosphere, or into a collection or a treatment system within the building or exterior to the building.
(3) In a typical facility, ventilation ducts are routed throughout the building. The ventilation ducts penetrate and cross fire separations, and typically comprise interior dampers installed within the fire separation section to prevent fire that penetrates the duct from travelling through the duet across the fire separators in the building. It will be appreciated that while such an implementation may be sufficient for the fire protection of ventilation ducts, ventilation or exhaust ducts for flammable or hazardous materials cannot be configured with fire dampers, so the duct itself must be fire-rated.
(4) To be classified as a fire-rated duct, an exhaust duct must be capable of preventing the release of flammable materials from inside the duct and/or combustible materials adjacent the exterior of the exhaust duct from catching fire if a fire exists on the other side of the duct. In other words, a

fire-rated duct must be capable of minimizing the transfer of heat through or across the duct walls. It is also desirable to maintain the wall thickness to a workable minimum.

(5) Fire-rated ducts are typically found in installations such as commercial kitchens and laboratories.

(6) In a commercial kitchen, the exhaust hoods are configured to capture grease laden air over deep fryers and grills, which is extremely flammable, and must be transported through the building to an exterior area where it can be safely discharged. Due to the flammable nature of the exhausted vapour, a minor fire, for example, in the kitchen could enter the exhaust duct and quickly spread throughout the duct system. As a result, any potential fire inside the duct system must be contained and thermal transfer through the duct walls limited to prevent ignition of adjacent combustible material in the kitchen or other areas of the building. In addition, the exhaust duct system must be capable of preventing the ignition of the grease laden air from a fire source in another part of the building and then spreading to the kitchen or other parts of the building where the exhaust duct system is routed.

(7) In a laboratory installation, the exhaust system is configured to collect and exhaust chemical vapours, including vapours from chemicals with low flash points, and contain any fire inside the duct system, or prevent an external fire from igniting the vapour inside the duct system.

(8) Known fire-rated exhaust duct systems are typically fabricated in sections, and the section are shipped to the installation location. At the installation location, the sections are welded together to form continuous conduits or conduit sections. Due to field conditions, the welding could be of poor work quality due to limited space and/or setup. This meant expensive rework and re-welding to seal leaks in the duct system during pressure testing. Conventional fire-rated duct systems typically required the installation of an additional gypsum fire-rated enclosure (approximately 10″ thick) around the duct. In addition to requiring an additional step, the gypsum enclosure was typically constructed/installed by another trade.

(9) In an attempt to overcome the known shortcomings in the art, chimney manufacturers introduced pre-fabricated fire-rated exhaust ducts based on a modification of existing chimney exhaust systems. While these pre-fabricated fire-rated exhaust ducts addressed shortcomings of existing systems, the characteristic round profile significantly limits the volume of air that can be vertically carried in conventional building footprints, and in a horizontal configuration, the round profile or cross section is often too large to fit into conventional ceiling space spaces or dimensions.

(10) Accordingly, there remains a need for improvements in the art.

BRIEF SUMMARY OF THE INVENTION

(11) The present invention comprises embodiments of a modular fire-rated duct system suitable for pre-fabrication and configured for assembly in the field.

(12) According to an embodiment, the present invention provides a modular fire-rated exhaust duct assembly comprising: two or more exhaust duct modules; each of said exhaust duct modules having an inner duct liner and an outer casing, and a void being formed between at least a portion of space between said inner duct liner and said outer casing, said void being configured for receiving an insulation material, and including one or more insulation encapsulation sections configured to connect to said inner duct liner and said outer casing to contain said insulation material in said void; a first exterior flange connector, and one end of each of said exhaust duct modules being configured for receiving said first exterior flange connector; a second exterior flange connector, and another end of each of said exhaust duct modules being configured for receiving said second exterior flange connector; said first and said second exterior flange connectors being configured to form a field assembly junction for coupling respective ends of said exhaust duct modules to form a single exhaust duct run; and a joint encasement section configured to be field connectable to each of said exhaust duct modules and encase said junction.

(13) According to another embodiment, there is provided an exhaust duct module configured to be

assembled in the field to form a fire-rated exhaust duct assembly, said exhaust duct module comprising: an inner duct liner formed with a generally rectangular cross-section; an outer casing formed with a generally rectangular cross-section and being sized to substantially surround said inner duct liner; a first flange connector configured to be attached to one end of said inner duct liner and one end of said outer casing, and couple said inner duct liner to said outer casing; a second flange connector configured to be attached to one end of said inner duct liner and one end of said outer casing, and couple said inner duct liner to said outer casing; and said first flange connector and said second flange connector being configured to be field attachable to couple another exhaust duct assembly.

(14) Other aspects and features according to the present application will become apparent to those ordinarily skilled in the art upon review of the following description of embodiments of the invention in conjunction with the accompanying figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Reference will now be made to the accompanying drawings which show, by way of example, embodiments according to the present application, and in which:

(2) FIG. **1** is a perspective view of an exhaust duct system according to an embodiment of the present invention and comprising a rectangular configuration and two sections or modules;

(3) FIG. **2** is an exploded view of one the sections of FIG. **1**;

(4) FIG. **3** is a longitudinal sectional view of the first section in FIG. **1** taken along line A-A;

(5) FIGS. **4**(a) to **4**(d) show alternate encapsulation configurations with varying thermal resistance characteristics according to embodiments of the present invention;

(6) FIGS. **5**(a) to **5**(c) show alternate configurations for finishing the incapsulation sections and flanges at the corners of the section according to embodiments of the present invention;

(7) FIG. **6** is a perspective view of an inner exhaust liner comprising first and second modular sections and the configuration for joining the two sections according to an embodiment of the present invention;

(8) FIG. **7** is a perspective of the inner exhaust liner of FIG. **6** showing a configuration for thermally encasing the mechanical joint between the first and second modular sections according to an embodiment of the present invention;

(9) FIG. **8** is a longitudinal sectional view of the inner exhaust liner of FIG. **7** taken through line B-B; and

(10) FIG. **9** is a partial and enlarged view of the joint encasement, i.e. the encased mechanical joint, for the inner exhaust liner of FIG. **7**.

(11) Like reference numerals indicate like or corresponding elements in the drawings.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(12) Reference is first made to FIG. **1**, which shows a fire-rated modular exhaust duct system or assembly according to an embodiment of the invention. The fire-rated modular exhaust duct is indicated generally by reference **100** and according to an embodiment comprises a plurality of exhaust duct sections or modules **110**, indicated individually by references **110**a and **110**b, as shown in FIG. **1**. The individual exhaust sections **110** are connected or coupled together with a mechanical joint indicated generally by reference **120** and described in more detail below. Embodiments of the modular fire-rated exhaust duct system or apparatus according to the present invention are suitable for installations or applications requiring fire-rated duct systems, such as exhaust duct systems for commercial applications, for instances, kitchens and restaurants, laboratories and other chemical or hazardous material processing facilities, as will be apparent to those skilled in the art.

(13) Reference is next made to FIG. **2**, which shows an exploded view of one of the exhaust duct modules or sections **110** according to an embodiment of the invention. The exhaust duct section **110** comprises an inner metallic duct **210**, first and second external flange connectors **220** (indicated individually by references **220**a and **220**b), insulation encapsulation sections or caps **230** (indicated individually by references **230**a, **230**b, **230**c, **230**d, **230**e, **230**f and **230**g), and an outer metallic casing **240**. According to an embodiment, the outer metallic casing **240** comprises separate metallic sheets or sections indicated individually by references **240**a, **240**b, **240**c and **240**d. According to another embodiment, one or more sheets or sections could be formed together as single pieces and then joined or connected together with the other section(s).
(14) As shown in FIGS. **2** and **4**, the inner metallic duct **210** is formed with a flange indicated generally by reference **212**. The flange **212** is configured to overlap a portion of the external flange connector **220**. According to an exemplary implementation, the inner metallic duct **210** is fabricated from stainless, carbon or coated steels, and is formed into a tube with a square or rectangular profile or cross-section. According to requirements under the UL 2221 and ASTM E2336 standards, the minimum wall thickness for the inner metallic duct **210** is typically 18 gauge (ga) for stainless steel and 16 ga for carbon steels.
(15) The inner metallic duct **210** can be fabricated or formed in a number of ways including: (1) forming a piece of metal into a tube; (2) forming two pieces of metal into "L" shaped sections and joining the two sections together to form a rectangle (or a square profile) tube; (3) forming a single piece of metal into a "U" shaped section and joining a flat piece of metal to the open end of the "U" shaped section; or (4) using four separate pieces or panels of steel and joining them to form a rectangular (or square) profile tube. The longitudinal joint or joints are continuously welded to provide a liquid and air tight seal between the edges of the panels. Other connection techniques, such as Pittsburgh type mechanical locks or pocket locks which are sealed with stitch welding can be utilized as will be understood by one skilled in the art. Such techniques can provide mechanical strength to pass the fire exposure tests.
(16) The flange **212** (FIGS. **2** and **4**) is formed at each transverse end of the inner metallic duct **210** and serves to define the duct section length. The flange **212** is formed at a right angle and provides a smooth flat surface for coupling adjacent duct sections and assisting in providing a liquid and air-tight surface, as described in more detail below.
(17) As shown in FIG. **4**(a) the inner duct liner **210** and the flange **212** are connected or affixed to the external flange connector **220**. The external flange connector **220** is fabricated from the same type of metal as the inner duct liner **210** so as to avoid the potential for galvanic action occurring between dissimilar metals. According to an embodiment, the external flange connector **220** is formed from four sections or pieces **222**, indicated individually by references **222**a, **222**b, **222**c and **222**d, in FIG. **2**. Each of the sections **222** comprises a flat metal strip formed into a right angle piece, or a section cut from hot rolled right angle structural steel. The sections **222** will typically have a thickness in the range of 0.125 to 0.250 inches to prevent the duct from collapsing from fire exposure. The thickness will also depend on the width and/or depth (i.e. length) of the duct section **110**. As shown, each of the flange sections **222** includes a series of holes **223** (punched or drilled), for example, at 4″ to 8″ centers to provide openings for receiving respective bolts **224** (FIG. **8**) which are secured by a respective nut **226** (FIG. **8**) to connect adjacent duct sections **110** together, as described in more detail below. As shown in FIGS. **5**(a) and **5**(b), the flange sections **222** can be connected using a square joint as indicated generally by references **510** and **520**, in order to assist in providing a liquid and air tight seal at the flange connection points for joining the duct sections **110**. Or as shown in FIG. **5**(c), the flange sections **222** can be connected utilizing a mitered joint as indicated generally by reference **530**, in order to assist in providing a liquid and air tight seal at the flange connection points for joining the duct sections **110**. According to an exemplary implementation, the external flange connector **220** is assembled around the flange end **212** of the inner duct **210** with the individual flange sections **222** being welded in place. For instance, the

flange sections **222** are welded to inner duct **210** with either a continuous weld or a stitch weld where the external flange connector **220**, i.e. the flange section **222**, interfaces with the flange or return end **212**, for example, at a point **410** and where the flange section interfaces with the inner duct **210** at a point **420**, as shown in FIGS. **4**(a) to **4**(d).

(18) Referring again to FIGS. **4**(a) to **4**(d), a space or void indicated by reference **430** is formed between the inner duct **210** and the outer metallic casing **240**. As shown, the void **430** can be filled with a thermally resistant insulation material or gas or liquid indicated generally by reference **440**. The thermal resistant material **440** is sealed or held inside the void **430** by the insulation encapsulation sections or caps **230** (**230**a, **230**b, **230**c and **230**d as depicted in FIG. **2**). The encapsulation sections or caps **230** are formed from flat metal pieces or strips having a thickness to provide sufficient strength to contain the insulation material in place, and also support the outer metallic casing **240** during exposure to fire. In a typical implementation, the encapsulation sections **230** have a thickness ranging from 16 ga to 24 ga.

(19) As shown in FIGS. **4**(a) to **4**(d) and further in FIG. **2**, the encapsulation sections **230** include respective flanges or returns indicated generally by reference **231**. The flanges **231** are formed along the longitudinal edges and provide attachment points or surfaces for connecting to the inner metallic duct **210** and the outer metallic casing **240**. For a rectangular or square duct configuration, the encapsulation sections or caps **230** comprise four sections **230**a, **230**b, **230**c and **230**d as depicted in FIG. **2**. The sections **230** can be joined using an overlap connection as shown in FIG. **5**(a), an overlap connection as shown in FIG. **5**(b) or a mitered connection as shown in FIG. **5**(c). The encapsulation sections **230** can have various cross-sections or profiles formed from solid or perforated material and will vary on the insulation material **440** being used and the required thickness of section **230**. As shown in FIG. **4**(a) and also FIG. **3**, the encapsulation section **230** comprises a flat or straight cross-section and is joined or connected to the inner duct **210** and the outer casing **240** utilizing and overlap or butt connection. The upper flange indicated by reference **231**a is connected or affixed to the outer casing **240** using a fastener **450**, for instance comprising a screw, a rivet or the like. For configurations where the encapsulation section **230** does not provide sufficient thermal resistance to prevent the outer casing **240** from exceeding thermal limitations according to test standards, an insulating gasket indicated by reference **452** can be installed on the face of the flange **231**a and thereby thermally separate the encapsulation section **230** from the outer casing **240** and provide a thermal bridge. The encapsulation section **230** is connected or affixed to the inner duct **210** by welding the lower flange **231**b to face or surface of the inner duct **210** by welding, for example, using stitch, continuous, seam or spot welding techniques. If the insulation **440** comprises a gas or a flowable material, then the seams between the encapsulation section **230**, the inner duct **210** and the outer casing **240** are sealed with a high temperature sealant **810** (FIG. **8**) to prevent the insulation material **440** from flowing from the space or void **430**.

(20) In addition to the straight or flat profile shown in FIG. **4**(a), the encapsulation section can comprise a stepped profile as indicated generally by reference **232** and shown in FIG. **4**(b), a saw tooth profile as indicated by reference **234** in FIG. **4**(c), or an accordion profile as indicated by reference **236** in FIG. **4**(d). By changing the profile of the encapsulation section **230**, the surface area and thermal properties can be varied and will be determined by the insulation material **440** being utilized. For a flat or straight profile, the encapsulation sections **230** can be joined using any one of the configurations shown in FIGS. **5**(a) to **5**(c). For a stepped profile **232**, the encapsulation sections **230** is joined using the mitered configuration shown in FIG. **5**(c). For the sawtooth profile **234** and the accordion profile **236**, the encapsulation sections **230** can be joined using the overlap connection configuration of FIG. **5**(a) or the mitered connection configuration of FIG. **5**(c).

(21) At the fabrication facility or factory, the order of assembly of the insulation encapsulation sections **230**, the inner metallic duct **210** and the outer metallic casing **240** will depend largely on the type of insulation material **440** being used to fill the void or space **430**. For instance, if the insulation material **440** comprises a blanket or batt type material, then the encapsulation sections

230 can be installed at both exterior flange connections **220** for a duct section **110** and the insulation can be friction fitted into place with the outer metallic casing **240** installed over the insulation **440**. If the insulation material **440** comprises a flowable material, but not a gaseous compound, the encapsulation sections **230** can be installed on one end of the exhaust duct section **110**, and the duct section **110** positioned vertically and the flowable insulation material **440** poured to fill the void **430** (FIG. **4**) and then the encapsulation sections **230** installed into place. If the insulation material **440** comprises a gaseous compound, then the encapsulation sections **230** are installed or connected to both ends of the exhaust duct section **110** and a sealant is also utilized to improve an air-tight seal in the void **430**. After the sealant has sufficiently cured, air in the void **430** is evacuated while at the same time injecting insulating gas, until the required concentration of insulating gas is achieved.

(22) Reference is made back to FIG. **1**. As shown in FIG. **1**, the two exhaust sections **110**a and **110**b are joined or coupled together by the mechanical joint **120**. According to an embodiment, the mechanical joint **120** comprises a joint encasement having a configuration as shown in FIGS. **7**, **8** and **9**, and is indicated generally by reference **700**. The joint encasement **700** is configured to encase or surround the joint formed between two adjacent exhaust duct sections **110**a and **110**b when the exterior flange connectors **220** are connected together, for example, bolted.

(23) As shown in FIG. **7**, joint encasement **710** comprises four sections **720**, indicated individually by references **720**a, **720**b, **720**c and **720**d. Each of the sections **720** comprises an outer metallic layer **730** and an insulating material (or layers of insulating material) **740**. The insulating material **740** is affixed to the inner face or side of the outer metallic layer **730** using suitable adhesives and/or mechanical fasteners, in order to provide for pre-fabrication at the factory and thereby minimize the number of components transported to and assembled on site. According to an embodiment, the outer metallic layer **730** comprises a flat piece of steel fabricated from the same type and gauge of metal used to fabricate the outer metallic casing **240** (FIG. **1**). As shown in FIGS. **8** and **9**, the outer metallic layer **730** has a width sufficient to overlap and extend past the fasteners **450** that connect the respective encapsulation sections **230** to the outer metallic casing **240**. The joint encasement sections **720** can comprise low profile configurations where the fasteners **450** comprise low profile fasteners, such as, rivets or spot welds. According to another aspect, the sections **720** having longitudinal edges formed with a safety or hemmed edge in order to stiffen the edge so that it lies flat against the outer metallic casing **240** when installed.

(24) As shown in FIGS. **8** and **9**, the joint encasement sections **720** comprise a hat configuration having a channel-like configuration with flanges or returns **760** formed outward from the legs of the channel to create an attachment surface or points for connecting to the outer metallic casing **240** using fasteners **722**, for example, screws, through holes (i.e. punched through holes) in the flanges **760**. According to another aspect, a thermal gasket indicated by reference **724** in FIG. **9** can be included to thermally isolate the joint encasement sections **720** from the outer metallic casing **240**.

(25) In accordance with an exemplary embodiment, the fire-rated exhaust duct sections **110** and encasement joint **120** is fabricated and assembled on a component level as described above in the factory and delivered to the installation site. According to an exemplary embodiment, a 0.250 to 0.500 inch diameter bead of high temperature sealant **810** is applied is applied to the face of one of the exterior flange connectors **220** adjacent the edge of the flange or return **212** (FIG. **4**(a)) on the inner metallic duct **210**. The second duct section **110**b is aligned with the first duct section **110**a and brought together so that the faces of the flanges **212** on both of the inner metallic ducts **210** are in contact. The bolts **224** (FIG. **6**) are then installed through each hole in the exterior flange connector **220** and the bolts **224** are secured by the nuts **226** (FIG. **6**). The fasteners are tightened in a circular and progressive manner to a torque setting defined by the exhaust duct manufacturer's specifications. Once the adjacent exhaust duct sections **110**a and **110**b are connected together, the encapsulation sections **230** are installed, one side at a time, until all four of the sections **230** have been installed. If the void **430** is to be filled with insulation **440**, the insulation **440** is put into place

before installation of the encapsulation sections **230**, as described above. Each of the joint encasement sections **720** is inserted into position and pressed down until the outer metallic layer **730** comes into contact with the surface of the outer metallic casing **240**. The joint encasement section **720** is secured by drilling holes through the flanges **760** and threading screws **722** into the outer metallic casing **240** as described above.
(26) In summary and according to another embodiment, the present invention comprises an exhaust duct system comprising a plurality of individual duct sections which are factory fabricated and then mechanically assembled on site. This eliminates the need to do on site fabrication of the duct sections, e.g. welding and other hot work. The exhaust sections are connected together to form longer sections and runs to create a fire-rated exhaust duct system in a building or other type of facility for exhausting or moving flammable or hazardous gases, vapours and materials from an originating source, e.g. an exhaust hood or another duct inlet, to a location where the flammable or hazardous gases, vapours or materials can be safely discharged, for example into the atmosphere, or into a collection or treatment system.
(27) The present invention may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. Certain adaptations and modifications of the invention will be obvious to those skilled in the art. Therefore, the presently discussed embodiments are considered to be illustrative and not restrictive, the scope of the invention being indicated by the appended claims rather than the foregoing description, and all changes which come within the meaning and range of equivalency of the claims are therefore intended to be embraced therein.

Claims

.[.1. A modular fire-rated exhaust duct assembly comprising: two or more exhaust duct modules; each of said exhaust duct modules having an inner duct liner and an outer casing, and a void being formed between at least a portion of space between said inner duct liner and said outer casing, said void being configured for receiving an insulation material, and including one or more insulation encapsulation sections configured to connect to said inner duct liner and said outer casing to contain said insulation material in said void; a first exterior flange connector, and one end of each of said exhaust duct modules being configured for receiving said first exterior flange connector; a second exterior flange connector, and another end of each of said exhaust duct modules being configured for receiving said second exterior flange connector; said inner duct liner comprising a flange configured to be affixed to said first exterior flange connector and a flange configured to be affixed to said second exterior flange connector; said first and said second exterior flange connectors being configured to form a field assembly junction for coupling respective ends of said exhaust duct modules to form a single exhaust duct run; and a joint encasement section configured to be field connectable to each of said exhaust duct modules and encase said junction..].
.[.2. The modular fire-rated exhaust duct assembly as claimed in claim 1, wherein said exhaust duct modules have a generally rectangular cross-section..].
.[.3. The modular fire-rated exhaust duct assembly as claimed in claim 1, wherein said inner duct liner and said outer casing are formed from a metallic material, and each having a thickness to provide a specified fire rating..].
.[.4. The modular fire-rated exhaust duct assembly as claimed in claim 1, wherein said insulation capsulation sections have a profile comprising one of a flat profile, a step profile, a sawtooth profile and an accordion profile..].
.[.5. The modular fire-rated exhaust duct assembly as claimed in claim 1, wherein said joint encasement section comprises a plurality of joint encasement sections, each of said joint encasement sections comprising an outer layer and an inner layer, and said outer layer having a flange on each longitudinal edge for affixing said encasement section to respective ends of said adjacent exhaust duct modules..].

.[.6. The modular fire-rated exhaust duct assembly as claimed in claim 5, wherein said outer casing is formed from a metallic material having a thickness sufficient to provide a specified fire rating and said outer layer of said joint encasement section comprises the same metallic material..].
.[.7. The modular fire-rated exhaust duct assembly as claimed in claim 6, wherein said joint encasement sections are configured to be field attachable to said respective surfaces of the said exhaust duct modules..].
.[.8. An exhaust duct module configured to be assembled in the field to form a fire-rated exhaust duct assembly, said exhaust duct module comprising: an inner duct liner formed with a generally rectangular cross-section, said inner duct liner having a flange at one end and a flange at another end; an outer casing formed with a generally rectangular cross-section and being sized to substantially surround said inner duct liner; a first flange connector configured to be attached to said flange at one end of said inner duct liner and one end of said outer casing, and couple said inner duct liner to said outer casing; a second flange connector configured to be attached to said flange at the other end of said inner duct liner and one end of said outer casing, and couple said inner duct liner to said outer casing; and said first flange connector and said second flange connector being configured to be field attachable to couple another exhaust duct assembly..].
.[.9. The exhaust duct module as claimed in claim 8, wherein a void is formed in at least a portion of a space separating said inner duct liner from said outer casing..].
.[.10. The exhaust duct module as claimed in claim 9, further including a plurality of insulation encapsulation sections configured to connect to said inner duct liner and said outer casing to close in said void space..].
.[.11. The exhaust duct module as claimed in claim 10, further including a thermal insulation material substantially filling said void space between said inner duct liner and said outer casing..].
.[.12. The exhaust duct module as claimed in claim 11, wherein said inner duct liner and said outer casing are formed from a metallic material, and each having a thickness to provide a specified fire rating..].
.Iadd.13. A modular fire-rated exhaust duct assembly comprising: two or more exhaust duct modules; each of said exhaust duct modules having an inner duct liner and an outer casing, and a void being formed between at least a portion of space between said inner duct liner and said outer casing, said void being configured for receiving an insulation material, and including one or more insulation encapsulation sections configured to connect to said inner duct liner and said outer casing to contain said insulation material in said void; a first exterior flange connector, and one end of each of said exhaust duct modules being configured for receiving said first exterior flange connector; a second exterior flange connector, and another end of each of said exhaust duct modules modules being configured for receiving said second exterior flange connector; said inner duct liner comprising a flange configured to be affixed to said first exterior flange connector and a flange configured to be affixed to said second exterior flange connector; said first and said second exterior flange connectors being configured to form a field assembly junction for coupling respective ends of said exhaust duct modules to form a single exhaust duct run; and at least one joint encasement section configured to be field connectable to each of said exhaust duct modules and encase said junction; wherein said at least one joint encasement section comprises a plurality of low-profile joint encasement sections, each of said low profile joint encasement sections having a flange formed on a longitudinal edge to form a longitudinal attachment surface, and said flange for each of said low-profile joint encasement sections comprising a low profile configuration for accommodating one or more low profile fasteners, and said longitudinal attachment surface for each of said low-profile joint encasement sections being configured to affix said encasement section to respective ends of adjacent said exhaust duct modules..Iaddend.
.Iadd.14. The modular fire-rated exhaust duct assembly as claimed in claim 13, wherein said exhaust duct modules have a generally rectangular cross-section..Iaddend.
.Iadd.15. The modular fire-rated exhaust duct assembly as claimed in claim 13, wherein said inner

duct liner and said outer casing are formed from a metallic material, and each having a thickness to provide a specified fire rating..Iaddend.
.Iadd.16. The modular fire-rated exhaust duct assembly as claimed in claim 13, wherein said insulation encapsulation sections have a profile comprising one of a flat profile, a step profile, a sawtooth profile and an accordian profile..Iaddend.
.Iadd.17. The modular fire-rated exhaust duct assembly as claimed in claim 13, wherein each of the said low-profile joint encasement sections comprises a hat configuration with a channel having an outwardly facing return formed on a longitudinal edge of said channel and each of said outwardly facing returns providing a respective attachment surface for affixing said low-profile joint encasement section to respective ends of said adjacent exhaust duct modules..Iaddend.
.Iadd.18. The modular fire-rated exhaust duct assembly as claimed in claim 17, wherein said outer casing is formed from a metallic material having a thickness sufficient to provide a specified fire rating and an outer layer of said joint encasement section comprises the same metallic material..Iaddend.
.Iadd.19. The modular fire-rated exhaust duct assembly as claimed in claim 18, wherein said joint encasement sections are configured to be field attachable to said respective attachment surfaces of the said exhaust duct modules..Iaddend.
